

Normal Big Bang Nucleosynthesis and the Primordial Lithium Problem

Hydrogen, helium and lithium production from standard-network controls to a qualified finite-temperature selective-transfer candidate

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Computational assistance and reproducibility support: ChatGPT Codex sol 5.5

EVIDENCE STATUS: Qualification-audit passed for the full Planck PR4/NPIPE CMB nuisance gate; scientific promotion remains conditional.

Executive Summary

- Standard BBN remains successful for deuterium and helium at the adopted CMB-compatible baryon density, but predicts Li-7/H roughly three to four times above the warm metal-poor stellar plateau.
- The July 2026 investigations were negative: FR/clock-proxy changes, SU2-style expansion controls, reaction-rate pulls and network scans did not produce a joint D/H, helium and lithium solution.
- The August 2026 programme identified a selective post-BBN electromagnetic-transfer candidate. The candidate is narrow and mechanism-dependent, but it has passed the abundance, cascade, entropy, FIRAS and leading finite-temperature checks recorded so far.
- The new September 2026 result is important: the combined native Planck PR4/NPIPE nuisance audit passes every preregistered gate. The registered lithium cell is inside the joint 95% CMB posterior with $\Delta\chi^2 = 1.257$.
- This does not prove the lithium problem is solved. It upgrades the evidence state from 'positive but ESS-limited' to 'CMB-gate audit-qualified'. Matched next-to-leading-order plasma physics, independent reproduction and an explicit particle/action-level model remain open.

TV index

Traceability item	Recorded value
Document update	Regenerated 11 September 2026 from local post-process and combined qualification audit outputs.
Authors	Clive Stewart Boyd; David Ng.
Computational support	ChatGPT Codex sol 5.5 referenced as the project computational-assistance environment.
Primary new gate	Full Planck PR4/NPIPE native nuisance audit: PASS.

Current Claim Boundary

The current result should be described as an audit-qualified candidate for a selective electromagnetic pathway through the cosmological lithium problem. The analysis does not yet constitute a discovery claim, a completed microscopic model, or a replacement for standard BBN. It does show that the registered August candidate is not excluded by the native Planck PR4/NPIPE CMB nuisance posterior under the applied gates.

Table 2. Evidence ladder status

Evidence layer	Status on 11 September 2026	Interpretation
Standard BBN reproduction	Pass	The ordinary D/H and helium success, plus lithium excess, was reproduced.
July global-modification controls	Fail for solution	Useful negative controls; no D+He+Li pass.
Selective mass-7 threshold window	Pass as sanity gate	Photons near the Be-7 threshold can affect mass-7 more selectively than late neutrons.
ACROPOLIS/LINX/FIRAS candidate	Pass at frozen candidate	Abundance and distortion gates are satisfied in the recorded candidate cell.
Full Planck nuisance CMB gate	Pass after ESS top-up	Combined chain passes split, ESS, registered density and nuisance stability gates.
Matched NLO plasma calculation	Open	Needed before strong physics promotion.
Independent reproduction	Open	Needed before external paper-strength claim.

Chronological Development and Test History

Table 1. Investigation chronology

Date	Investigation stage	Result status
16-17 July 2026	Standard BBN, FR/clock-proxy, neutron-lifetime, reaction-rate and SU2-style expansion controls with LINX validation.	Negative result: no joint D/H, helium and Li-7 solution. This is the explicit failure boundary for the preliminary July programme.
17-18 July 2026	Hierarchical and surrogate follow-up tests, plus strict matrix validation.	Useful for scoping but not a solution. Lithium could be moved only at the cost of other abundance or CMB gates.
20-21 August 2026	Selective post-BBN electromagnetic-transfer hypothesis. Threshold screen, ACROPOLIS	Positive candidate emerged: a narrow MeV-scale window can reduce mass-7 while

	cascade tests, frozen LINX priors and production modelling.	preserving D/H, Y_p , He3/D, entropy and FIRAS gates.
21 August 2026	Full Planck PR4/NPIPE CMB nuisance-likelihood protocol launched using Cobaya/CAMB and native Planck likelihoods.	Initial status pending. The goal was to replace a compressed CMB Gaussian gate with a native spectrum-level posterior.
2 September 2026	Main full CMB nuisance chain completed and post-processed.	Positive but not yet fully qualified: Cobaya convergence was reported, but the stricter independent audit found autocorrelation ESS below the preregistered 1000 gate for some nuisance parameters.
2-11 September 2026	ESS top-up chain run after the main chain to strengthen nuisance-tail sampling.	Computer reset stopped the top-up, but not before adding enough samples for the combined audit.
11 September 2026	Combined main plus ESS top-up qualification audit.	Pass: all preregistered audit gates passed, including split-chain stability, autocorrelation-aware ESS, nuisance stability, half-chain stability and registered-cell posterior density.

TV index

Traceability item	Recorded value
Chronology source	July 2026 local BBN/Li outputs, August full CMB nuisance folder, September combined qualification audit.
Failure boundary	July preliminary programme failed to solve lithium while preserving D/H, helium and CMB-compatible baryon density.
Positive boundary	August selective-transfer candidate became positive only after moving from global expansion/clock controls to a narrow post-BBN electromagnetic mechanism.

Scientific Background

BBN predicts light-element abundances from the baryon density, the radiation content, neutron-proton conversion, nuclear reaction rates and the expansion history. The same framework that gives successful deuterium and helium predictions leaves a persistent Li-7 excess when compared with warm metal-poor halo-star plateaux. Most primordial Li-7 is produced first as Be-7, so a selective mechanism can target mass-7 after ordinary nucleosynthesis without needing to disrupt all light nuclei.

Core Equations

Table 3. Formula cross-reference

Label	Expression	Use
E1	$\eta_{10} = 273.78 \, \omega_b$	CMB-to-BBN baryon-density conversion for $T_{\text{CMB}} = 2.7255 \text{ K}$.
E2	$dY_i/dt = \sum_j N_{ij} \Gamma_{ij}(Y,T)$	Network abundance evolution in schematic form.
E3	$\text{Li discrepancy factor} = (\text{Li-7/H})_{\text{BBN}} / (\text{Li-7/H})_{\text{plateau}}$	Measures the anomaly scale.
E4	$E_{\gamma} \text{ approximately } m_{\phi} / 2$	Two-photon decay energy for the mediator candidate.
E5	$\Delta \chi^2 = (x - \mu)^T C^{-1} (x - \mu)$	Registered-cell density test in the CMB posterior.
E6	$C = L L^T; C^{-1} = L^{-T} L^{-1}$	SPD covariance cache used for the Planck PR4/NPIPE likelihood.

TV index

Traceability item	Recorded value
Physics boundary	Equations are diagnostic and model-selection support. A full Lagrangian/action is still required for theory-level promotion.
Symmetry boundary	The BBN-Li mediator model is not currently claimed as a Noether-derived symmetry result.

August Candidate Cell

The August candidate moved away from broad changes to the expansion law and toward a selective post-BBN transfer. Its working idea is simple: an electromagnetic injection can be tuned to interact with mass-7, principally Be-7, more efficiently than with deuterium, provided the spectrum and timing remain narrow enough.

Table 4. ACROPOLIS/LINX candidate and CMB comparison

Quantity	Value
mediator_mass_MeV	4.44
photon_energy_MeV	2.22
lifetime_s	1.584893e+05
n_phi_over_n_gamma_at_10MeV	1.211528e-06
g_phi_gamma_GeV_inv	9.767274e-11
D_H_times_1e5	2.493161
Yp	0.246735
Li7_H_times_1e10	1.488554
He3_D	0.415242
eta10_CMB	6.096517
Neff_CMB	3.044601
mu_distortion	1.476434e-09
joint_pass_probability	0.650158
delta_log_evidence_vs_standard	105.206012
Planck bestfit eta10	6.105374
Planck bestfit nnu	2.84265
Planck bestfit YHe	0.282034
Planck bestfit chi2_CMB	10971.285

TV index

Traceability item	Recorded value
Candidate source	Post-process report and local resummed ACROPOLIS/LINX candidate values.
Abundance anchor	Li-7/H = 1.488554e-10 in candidate units table, with D/H and Y_p retained inside gates.
Spectral-distortion anchor	mu_distortion = 1.476434e-09, below the adopted FIRAS-style gate.

Full Planck PR4/NPIPE CMB Nuisance Gate

The full CMB gate replaced the earlier compressed Gaussian approximation with a native spectrum-level Planck likelihood. It samples eta10 through omega_b, samples N_eff and Y_p independently, and includes the Planck high-l CamSpec TTTEEE likelihood, low-l TT, low-l EE and PR4 lensing, plus foreground and calibration nuisance parameters.

Table 5. Planck likelihood stack

Component	Role
Planck PR4/NPIPE CamSpec TTTEEE	High-multipole CMB spectra and covariance.
Planck 2018 low-l TT	Large-scale temperature likelihood.
Planck 2018 low-l EE	Large-scale polarisation likelihood.
Planck PR4 lensing	Lensing reconstruction constraint.
Foreground/calibration nuisance block	Native amplitudes, tilts and calibration parameters left free under their likelihood priors.

Table 6. Main full CMB nuisance posterior summary

Parameter	Mean	Std	Median	68% interval	95% interval
ombh2	0.02219	1.9507e-04	0.02219	0.021995 to 0.022381	0.021809 to 0.022573
nnu	2.81855	0.2532	2.81194	2.569554 to 3.062147	2.343549 to 3.327882
YHe	0.262427	0.0186	0.262668	0.244057 to 0.280933	0.225078 to 0.298364
eta10	6.075084	0.0534	6.075231	6.021877 to 6.127459	5.97098 to 6.179927
H0	66.210921	1.5058	66.164438	64.706901 to 67.68329	63.421067 to 69.23348

omegam	0.31734	0.0087	0.317346	0.308729 to 0.326008	0.300514 to 0.334223
sigma8	0.803256	0.0095	0.803151	0.793847 to 0.812604	0.784934 to 0.822145
S8	0.826006	0.0111	0.826026	0.814952 to 0.837015	0.804369 to 0.847849
chi2__CMB	10984.919918	7.9961	10984.146	10979.26564 to 10990.31448	10975.553 to 10997.62775

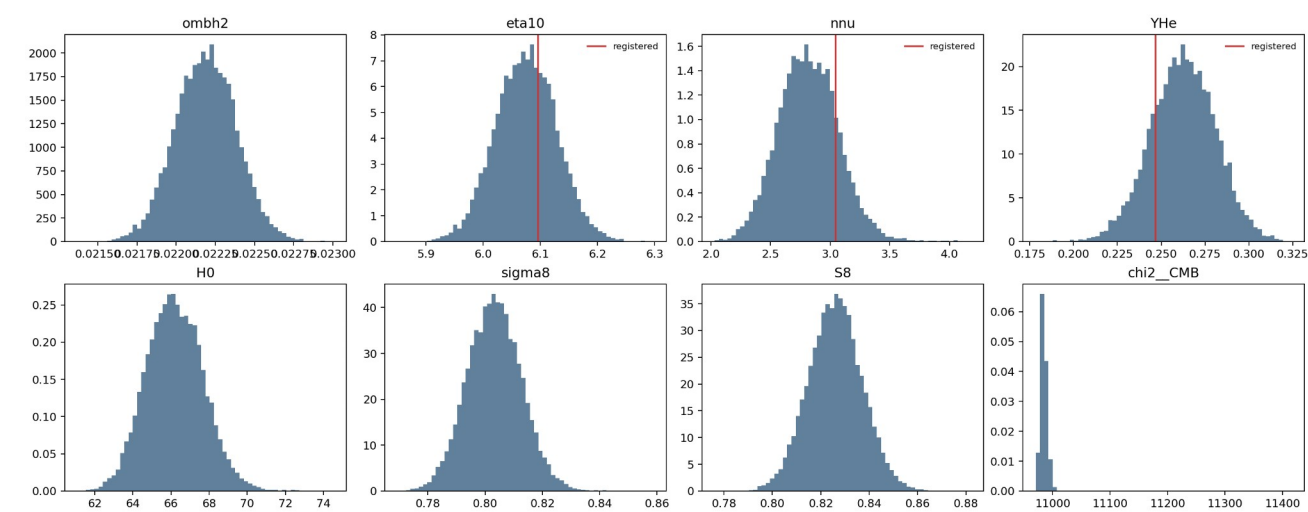


Figure 1. Key posterior summaries from the Planck PR4/NPIPE nuisance run.

TV index

Traceability item	Recorded value
Protocol	C:\Users\clive\Documents\Codex\2026-07-14\ok-contin\outputs\bbn_lithium_full_cmb_nuisance_20260821\full_cmb_nuisance_protocol_2026-08-21.md
Post-process	C:\Users\clive\Documents\Codex\2026-07-14\ok-contin\outputs\bbn_lithium_full_cmb_nuisance_20260821\postprocess\planck_pr4_full_nuisance_postprocess_report.md
Native-likelihood caveat	This is not a Planck map-level reanalysis and does not include ACT or SPT.

September Qualification Audit

The September audit combines the completed main chain with the ESS top-up chain. It uses a stricter gate set than the initial post-process: split-chain posterior summaries, autocorrelation-aware ESS, registered-cell posterior density, nuisance-parameter stability, all-row versus last-half comparison and explicit pass/fail scoring.

Table 7. Preregistered qualification gates

Gate	Threshold	Observed	Pass
Cobaya split-chain mean R-1	<= 0.03	0.0293	True
Independent split R-1 max over ombh2, nnu, YHe, nuisance	<= 0.03	0.0021	True
Autocorrelation ESS min over ombh2, nnu, YHe, nuisance	>= 1000	1449.0004	True
Registered lithium cell joint posterior density	delta-chi2 <= 7.814728	1.2574	True
Nuisance split stability diagnostic	max split mean shift <= 1 sigma	0.0838	True
First-half vs second-half stability diagnostic	max mean shift <= 1 sigma	0.0926	True

Table 8. Selected audit metrics

Parameter	Split R-1	Autocorr ESS	tau_int	Half-shift sigma	Status
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Table 9. Registered lithium cell in the combined CMB posterior

Quantity	Value
posterior mean eta10	6.074645
posterior mean N_eff	2.814036
posterior mean Y_p	0.262997
registered eta10	6.096365
registered N_eff	3.044709
registered Y_p	0.246733
registered Delta chi2	1.25743
Mahalanobis distance	1.121352
density ratio to posterior peak	0.533277
posterior mass inside equal-density contour	0.261374
joint 95 gate pass	True
weighted kNN density ratio	0.601247

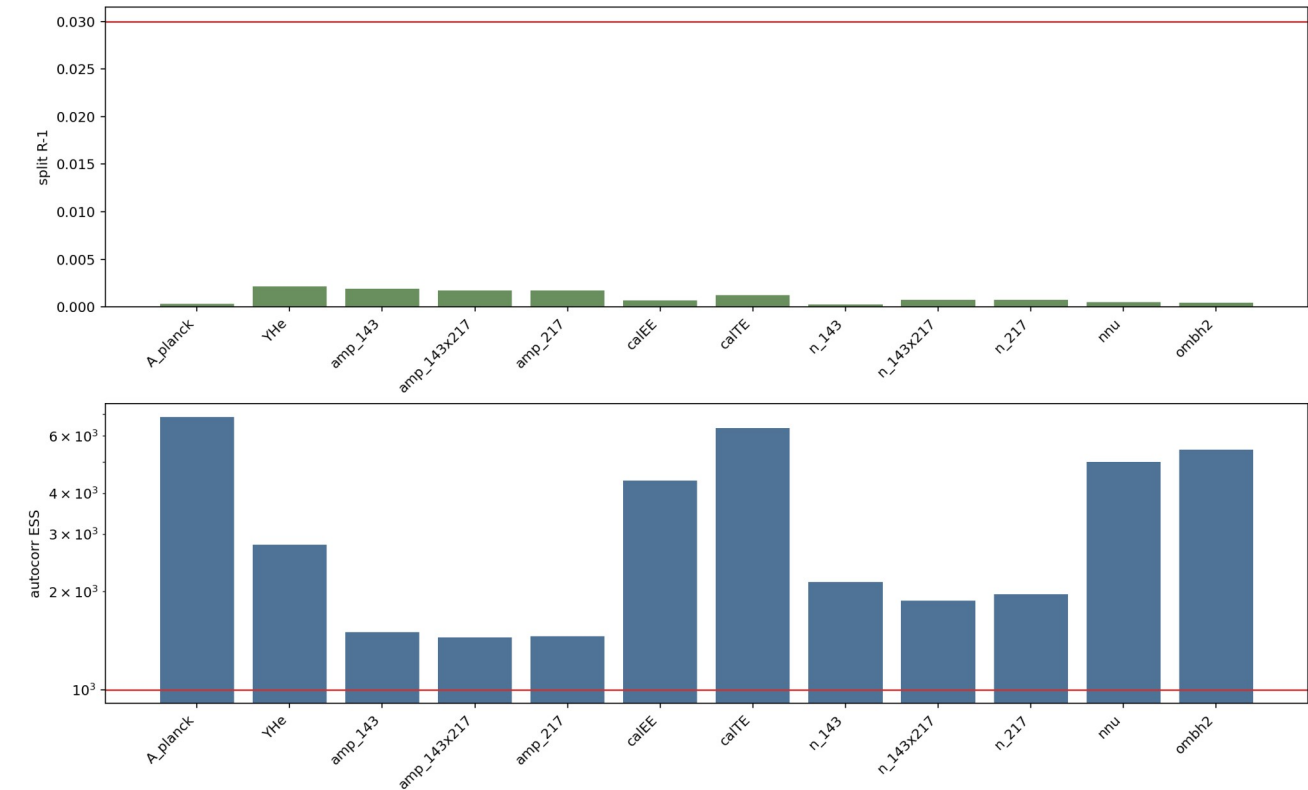


Figure 2. Qualification gate readout for the combined main plus ESS top-up chain.

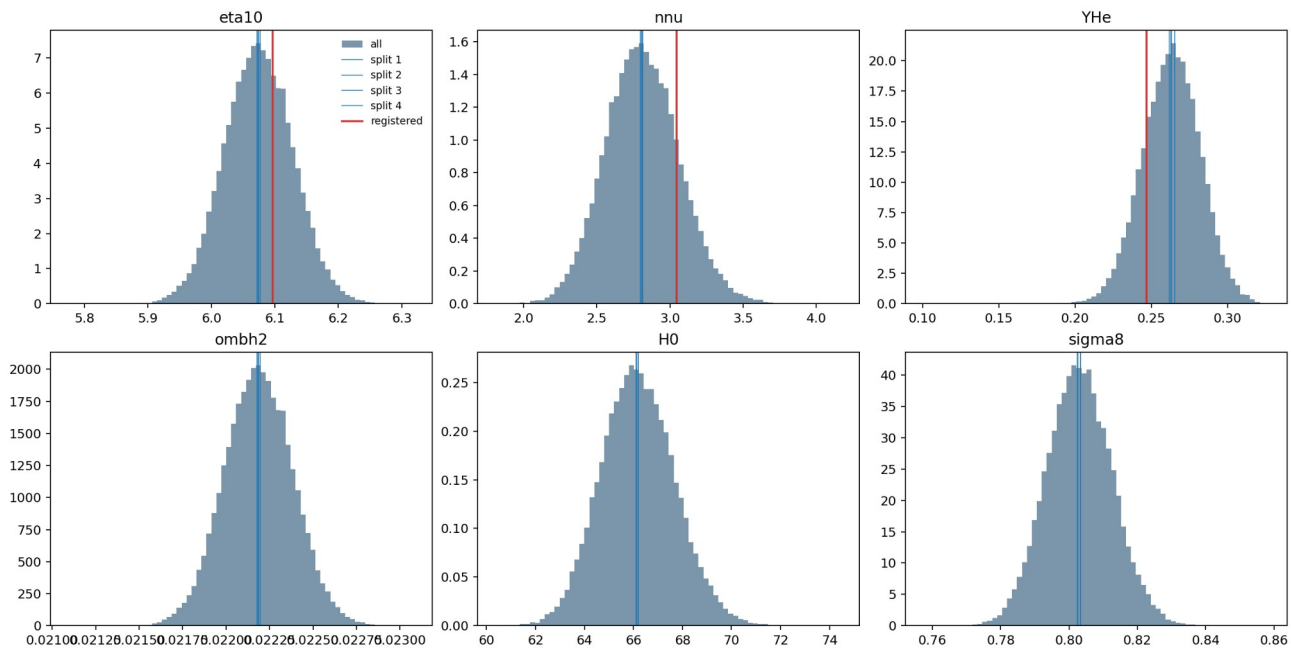


Figure 3. Split-chain posterior overlay used to audit stability.

TV index

Traceability item	Recorded value
Combined chain	C:\Users\clive\Documents\Codex\2026-07-14\ok-contin\outputs\bbn_lithium_full_cmb_nuisance_20260821\chains_combined\planck_pr4_full_nuisance_main_plus_ess_topup.1.txt
Audit report	C:\Users\clive\Documents\Codex\2026-07-14\ok-contin\outputs\bbn_lithium_full_cmb_nuisance_20260821\qualification_audit_combined_ess_topup\planck_pr4_full_nuisance_qualification_audit.md
Headline	overall_pass=True; min autocorrelation ESS=1449; max split R-1=0.002143; registered Delta chi2=1.257.
Caveat	Cobaya log R-1 line is inherited from the main run log; independent split R-1 and ESS are recomputed from the combined chain.

Interpretation

The main interpretation has changed since 28 August 2026. At that stage the Planck nuisance gate was still pending. After the main chain completed on 2 September 2026 it was positive but not fully qualified because the stricter autocorrelation-aware ESS gate failed for some nuisance directions. The ESS top-up, combined with the main chain and audited on 11 September 2026, removes that immediate technical objection.

The data therefore support the statement that the registered lithium candidate lies in an acceptable part of the native Planck PR4/NPIPE posterior for eta10, N_eff and Y_p. This is narrower than claiming the mediator exists. It is nevertheless materially stronger than the August status because it passes the spectrum-level CMB nuisance treatment rather than relying on a compressed CMB approximation.

Future Research and Test Options

1. Matched next-to-leading-order thermal calculation for the same mediator point, including pole-cut consistency and thermal self-energy treatment.
2. Independent reproduction of the ACROPOLIS/LINX abundance chain and the Planck nuisance posterior audit on a separate machine or environment.
3. Sensitivity analysis over nuclear-rate choices, stellar plateau assumptions, D/H observational anchors and helium determinations.
4. Explicit action-level particle model: specify the field content, Lagrangian, production mechanism, decay channel, symmetries and conservation laws.

- 5. External CMB and spectral tests: ACT/SPT consistency, improved FIRAS-style spectral-distortion treatment and future PIXIE-like sensitivity projection.
- 6. Publication-grade reproducibility package: frozen manifests, file hashes, environment locks, CSV tables and figure regeneration commands.

Appendix A. Acronyms

Table A1. Acronym glossary

Acronym	Meaning
ACROPOLIS	Public electromagnetic-cascade framework used for BBN photodisintegration constraints.
AIC	Akaike Information Criterion; lower is preferred after a penalty of 2 parameters.
BBN	Big Bang Nucleosynthesis.
BIC	Bayesian Information Criterion; lower is preferred with a stronger log(N) parameter penalty.
CAMB	Code for Anisotropies in the Microwave Background.
CMB	Cosmic microwave background.
Cobaya	Cosmological Bayesian analysis package used for the Planck nuisance MCMC.
D/H	Deuterium-to-hydrogen abundance ratio.
ESS	Effective sample size; here the autocorrelation-aware number of effectively independent posterior samples.
FIRAS	Far Infrared Absolute Spectrophotometer on COBE.
LINX	Fast differentiable BBN package used for abundance calculations and validation.
MCMC	Markov-chain Monte Carlo.
NPIPE	Planck reprocessing pipeline used for PR4 products.
PR4	Planck public release 4.
SPD	Symmetric positive definite; used for covariance handling and Cholesky/LAPACK inversion.
TV index	Traceability/validation index. The reports place it after each investigation section.

Appendix B. Variables

Table B1. Variable glossary

Variable	Meaning
eta10	10^10 times the baryon-to-photon ratio eta; converted from omega_b using eta10 = 273.78 omega_b for T_CMB = 2.7255 K.
omega_b / ombh2	Physical baryon density Omega_b h^2.
omega_c / omch2	Physical cold dark matter density Omega_c h^2.
N_eff / nnu	Effective relativistic species count inferred independently in the CMB nuisance chain.
Y_p / YHe	Primordial helium-4 mass fraction; sampled independently in the full CMB likelihood.
D/H	Deuterium abundance anchor; required to remain within the adopted observational gate.
Li-7/H	Lithium-7 abundance. Most primordial mass-7 is produced as Be-7 and later decays to Li-7.
m_phi	Mediator mass in MeV for the selective electromagnetic-transfer candidate.
E_gamma	Characteristic photon energy, approximated as m_phi/2 for two-photon decay.
tau_phi	Mediator lifetime in seconds.
n_phi/n_gamma	Mediator abundance relative to photons at T = 10 MeV.
g_phi_gamma	Effective mediator-photon coupling in GeV^-1.
mu	CMB spectral-distortion chemical-potential style readout.
R-1	Convergence diagnostic used by Cobaya and the split-chain audit.
chi2	Goodness-of-fit statistic; lower values indicate a better fit for the same data and model complexity.

Appendix C. Program and Data Manifest

Table C1. Principal programs

Program or path	Role in the investigation
analyze_bbn_lithium_linx_fr_network_2026-07-16.py	July LINX/FR network controls; reproduced the lithium problem and did not identify a D+He+Li pass.
analyze_su2_bbn_lithium_gate_2026-07-17.py	July SU2-style expansion/thermal gate for lithium; part of the negative control phase.
validate_bbn_lithium_linx_matrix_2026-07-17.py	Strict July validation matrix for the BBN lithium controls.
audit_bbn_lithium_linx_fr_scan_2026-07-17.py	Audit layer for the July LINX/FR scan.
run_cobaya_spd_cache_builder_2026_08_21.py	Built the validated SPD cache for the Planck PR4/NPIPE CamSpec likelihood.
postprocess_planck_pr4_full_nuisance.py	Post-processed the native full CMB nuisance chain and compared the registered lithium cell with the CMB posterior.
audit_planck_pr4_full_nuisance_qualification.py	Strict qualification audit: split chains, autocorrelation-aware ESS, registered-cell density, nuisance stability and half-chain stability.
start_planck_pr4_full_nuisance_background.ps1	Background launcher for the native Planck PR4 full nuisance MCMC.
start_planck_pr4_full_nuisance_ess_topup_background.ps1	Background launcher for the ESS top-up chain.
build_bbn_li_reports_20260911.py	This reproducible DOCX report generator.

Table C2. Repository and output cross-reference

Object	Path or URL
Local output root	C:\Users\clive\Documents\Codex\2026-07-14\ok-continuu\outputs\bbn_lithium_full_cmb_nuisance_20260821
Combined qualification audit	C:\Users\clive\Documents\Codex\2026-07-14\ok-continuu\outputs\bbn_lithium_full_cmb_nuisance_20260821\qualification_audit_combined_ess_topup
Post-process folder	C:\Users\clive\Documents\Codex\2026-07-14\ok-continuu\outputs\bbn_lithium_full_cmb_nuisance_20260821\postprocess
GitHub project	https://github.com/cliveboyd/Cosmology
GitHub BBN code	https://github.com/cliveboyd/Cosmology/tree/main/github_export/code/bbn_lithium
GitHub papers	https://github.com/cliveboyd/Cosmology/tree/main/github_export/docs/papers

Appendix D. References

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